Embedded and Edge AI for UWB Time-Series

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I. Introduction

Advancements in neural network design and hardware co-optimization have laid down the foundations for the field of Embedded and Edge AI. Intelligence (*i.e. predictive modeling*) can be deployed right on sensing grounds, balancing responsiveness and accuracy with model size and device performance. Such developments prove to be crucial for applications handling sensitive data, as information can be processed exclusively on-device, and real-time tasks where the delay of a cloud infrastructure would be incapacitating. This report presents a localization and crowd estimation Deep Learning model developed for a low-end Micro-Controller Unit (MCU), a fitting use case for a local-first, privacy driven application.

A. Problem Setting and Constraints

The desired outcome is to estimate, at each time frame, the number of people within a 4.8×7.2 meters indoor environment and their two-dimensional positions. The available data are measurements from six synchronized Ultra-Wideband (UWB) radar boards [1] arranged as displayed in Fig. 1a. Each board contains three spatially separated antennas, allowing the model to exploit CIR magnitude, inter-antenna phase differences, and their temporal evolution.

The model must return an unordered set of *zero to four* $((x, y))$ coordinates in meters. Performance on the localization task is evaluated through per-frame Hungarian matching, which finds the one-to-one assignment with minimum total Euclidean distance. Following the minimum-cost one-to-one Hungarian assignment, a prediction counts as correct if the Euclidean distance from the assigned ground-truth position does not exceed 1 m.

The final TensorFlow Lite model must respect the ESP32-S3 deployment limits of less than *800KB* of model storage and less than *300KB* of activation memory, including pre-processing tasks.

B. Main Contributions

The proposed system combines a convolutional antenna encoder, a global fully connected bottleneck, and a CenterNet-inspired heatmap head [2] trained on pre-processed windows. Its full-INT8 model yields 0.9751 F1 at 1-m Hungarian-matching radius on held-out windows `3` and `11`, with 0.1391 m median matched error. Storage amounts to 521.52 KiB, with a 159.19 KiB estimated activation arena. A real-time variant based on live EMA background removal obtains 0.9498 F1 under the same data split, without the 80 ms lookahead delay.

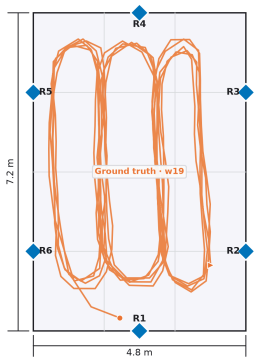


Fig. 1a. Room setup for the six radars. Trajectory from W19.

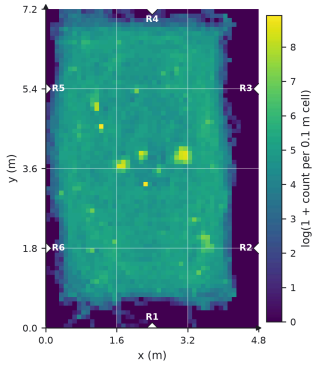


Fig. 1b. Spatial coverage measured over the complete dataset.

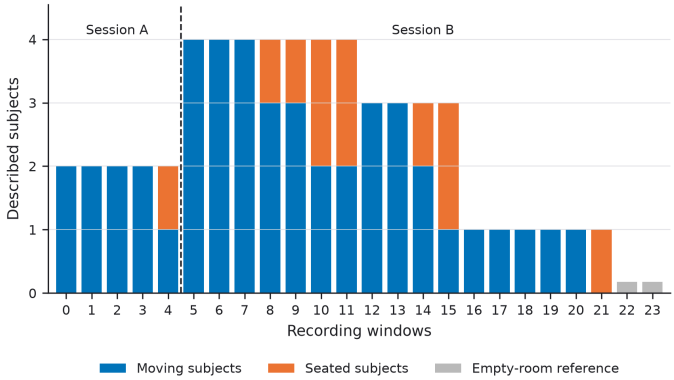


Fig. 2. Subject behaviours across acquisition windows.

II. Dataset

The dataset [3] contains 24 five-minute acquisitions. At a sampling rate of 25 Hz, this amounts to $180,000$ frames. Data were recorded in two sessions with different sensor calibrations. Recorded behaviours include single- and multi-subject random walks (19 instances), seated subjects (7), serpentine walks (2), and empty-room recordings (2, both from the second session); several acquisitions contain more than one behaviour (Fig. 2). Fig. 1b shows the aggregate spatial coverage. Each antenna provides 120 range bins of 0.15 m, spanning a nominal range of 18 m, while the targets lie on the $4.8\text{m} \times 7.2\text{m}$ room plane.

A. Input features

Given six boards, three antennas per board, 120 range bins, and the in-phase and quadrature components $(I, Q) \in \mathbb{R}^2$ of the complex CIR, the observation at time t is:

$$\mathbf{x}_t \in \mathbb{R}^{6\times3\times120\times2} \tag{1}$$

B. Target variables

The corresponding targets are

$$\mathbf{p}_t \in \mathbb{R}^{4\times 2} \quad \text{and} \quad \mathbf{m}_t \in \{0,1\}^4, \tag{2}$$

where $\mathbf{p}_t$ contains up to four 2D positions and $\mathbf{m}_t$ is the valid entry mask. The number of people present in frame t is consequently

$$M_t = \sum_{i=1}^{4} m_{t,i}. \tag{3}$$

III. Pre-processing techniques

A. Background noise removal

We propose two versions of the same architecture. The most accurate model uses complete-acquisition statistics and centered temporal context, therefore being non-causal. The second is a *causal* model: predictions and pre-processing state depend only on measurements available up to the current frame. The heatmap formulation is unchanged, but differ in background estimate, normalization, and temporal ordering.

Consistent with previous UWB literature [4], both versions suppress the static CIR component before computing magnitude and inter-antenna phase features. Empty-room windows `22–23` provide a potential common reference, but they were acquired in a different session from windows `0–4`. We quantified this mismatch using:

$$\rho_w = \frac{\sum \left| \text{mean}_{t(X_{w(t)})} - B_{\text{empty}} \right|}{\sum \left| \text{mean}_{t(X_{w(t)})} \right|}, \tag{4}$$

where $X_{w(t)}$ is the complex CIR of window w and B_{empty} is the temporal mean pooled from windows `22–23`. The resulting ratio was approximately 0.917 for windows `0–4`, compared with a median of 0.047 across the non-empty windows `5–21`. The same empty room reference can remove most static clutter from windows `5–21`, but cannot be assumed to represent windows `0–4`.

a) *Own-mean background (most accurate):*

The submitted pipeline computes the complex temporal mean of the *entire* acquisition for every frame, adapting to clutter/noise that is acquisition-specific, at the cost of potentially masking a person who remains stationary for most or all of an acquisition. In a single seed comparison, no subtraction, fixed-reference subtraction, and own-mean subtraction achieved test F1 scores of 0.537, 0.780, and 0.873, with precision/recall of $(0.898, 0.382)$, $(0.843, 0.725)$, and $(0.958, 0.802)$, respectively. Own-mean subtraction also led on validation (0.952 versus 0.879 and 0.666), but failed on fully static window `21` (0.075 F1 versus 0.940 for the fixed reference). We select it as best performing, albeit a clear stationary target limitation.

b) *Causal EMA background:*

The online version maintains a complex background estimate B_t. Features for frame t are computed from the *residual* $X_t - B_t$ *before* the state is updated. An update is enabled after 25 consecutive frames whose residual ratio is no greater than 0.08. The background then advances according to a *clipped* exponential moving average (EMA),

$$B_{t+1} = B_t + \alpha \, \text{clip}(X_t - B_t), \tag{5}$$

where $\alpha = 0.001$ and each complex update is limited to 5% of the magnitude of the current background coefficient. Gate and clipping limits reduce the risk of absorbing useful information into the background. The state can be initialized from an empty room estimate; at the first frame, a mismatch ratio greater than 0.3 causes reinitialization of the estimate. Without external references, we simply use the first frame, again at the cost of missing a stationary target.

B. Bin pruning

For each frame $\mathbf{x}$, the complex CIR is reconstructed from its in-phase and quadrature components. Bins `0–4`, dominated by direct `TX→RX` coupling, and bins `35–119`, representing ranges beyond approximately 5.25 m, are discarded. The retained interval, bins `5–34`, contains 30 of the original 120 bins and reduces the range-bin dimension by 75%.

C. Log-normalization and phase differences

After background subtraction, both variants normalize the residual magnitude using $\log(1 + |x|)$. The main variant's pipeline divides magnitude by the 99th percentile computed over the complete acquisition. The causal pipeline instead uses the fixed scale 6.511169 derived from training, and the median of the 99th percentile log-residual magnitudes in the training data, avoiding run-time scale drift. In both variants phase differences are formed for each board, for antenna pairs $(0,1)$, $(1,2)$, and $(0,2)$. The channel order is (normalized) amplitude, cosine, then sine, producing one feature frame of shape $18 \times 30 \times 3$.

D. Consecutive frames

The submitted network consumes $K = 5$ frames centered on the frame being predicted, $(t-2, ..., t+2)$, with zero padding where the centered window extends beyond an acquisition. At 25 Hz, the two future frames introduce 80 ms of algorithmic look-ahead. The online variant instead stacks $(t-4, ..., t)$ and zero pads unavailable startup history.

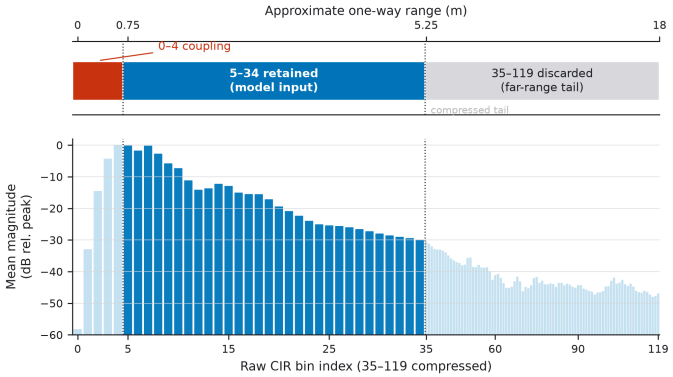


Fig. 3. Magnitude mean and variance of CIR frames across the entire dataset.

IV. Deep-learning techniques

The proposed network adapts the anchor free CenterNet formulation of Zhou et al. [2] to UWB-signal person localization. Instead of predicting bounding boxes from images, the model learns to predict a *coordinate heatmap*, where *local maxima* indicate predicted positions, given a UWB time-window. The architecture consists of an independent encoder for each radar, a global sensor-to-room bottleneck, a room-to-coordinate decoder, and a final heatmap head.

A. Antenna Encoder

The encoder combines local temporal, range, and intra-radar antenna information. A 1×1 Conv2D stem maps the 15 channels of the $18 \times 30 \times 15$ input to 24 channels at each antenna and range bin. Two depthwise-separable 1×3 blocks then operate along the range axis and are followed by range-only max-pooling. A third 3×1 block with stride 3 combines the three antennas of each radar while preventing mixing between adjacent radars. Batch normalization and ReLU follow the convolutions, yielding a $6 \times 3 \times 96$ tensor.

Convolution kernels are *shared* across spatial locations, and depthwise-separable convolution reduces parameters by separating the spatial mixing step and channel mixing step.

B. Global bottleneck

After a 1×1 Conv2D module halves the channel count, the $6 \times 3 \times 48$ representation is flattened. A fully connected layer maps it to a 256-dimensional global room representation in which information from all radars is fused. Dropout and L_2 regularization are active only during training and target the dense layers, which contain most of the model parameters.

C. Dense Projection and Convolutional Decoder

The bottleneck is projected to an 864-dimensional vector and reshaped to a 9×6 room grid with 16 channels ($9 \times 6 \times 16 = 864$). This operation reintroduces spatial structure. The convolutional decoder then imposes a local smoothness prior consistent with the Gaussian targets in Section IV.D.

The decoder comprises two upsampling stages. Each applies $2\times$ nearest-neighbour upsampling followed by two 3×3 convolutional blocks with batch normalization and ReLU. Resolution increases while channel count decreases, producing a final $36 \times 24 \times 12$ tensor. Compared with a direct full-resolution dense projection, this design reduces dense-layer cost and lets weight-shared convolutions refine the room representation before heatmap prediction.

D. Heatmap Head

The final head transforms the decoder output into the set-valued localization required by the task. A 3×3 convolution reduces the $36 \times 24 \times 12$ room representation to a *1-channel logit map*, followed by a *sigmoid activation*:

$$\boldsymbol{h}_t \in \mathbb{R}^{36\times24\times12} \longrightarrow \hat{\boldsymbol{Y}}_t \in [0,1]^{36\times24}. \tag{6}$$

The independent sigmoid maps each cell to a confidence score, allowing multiple centers (or none) to be active simultaneously. Each output cell represents a $0.2\text{m} \times 0.2\text{m}$ region of

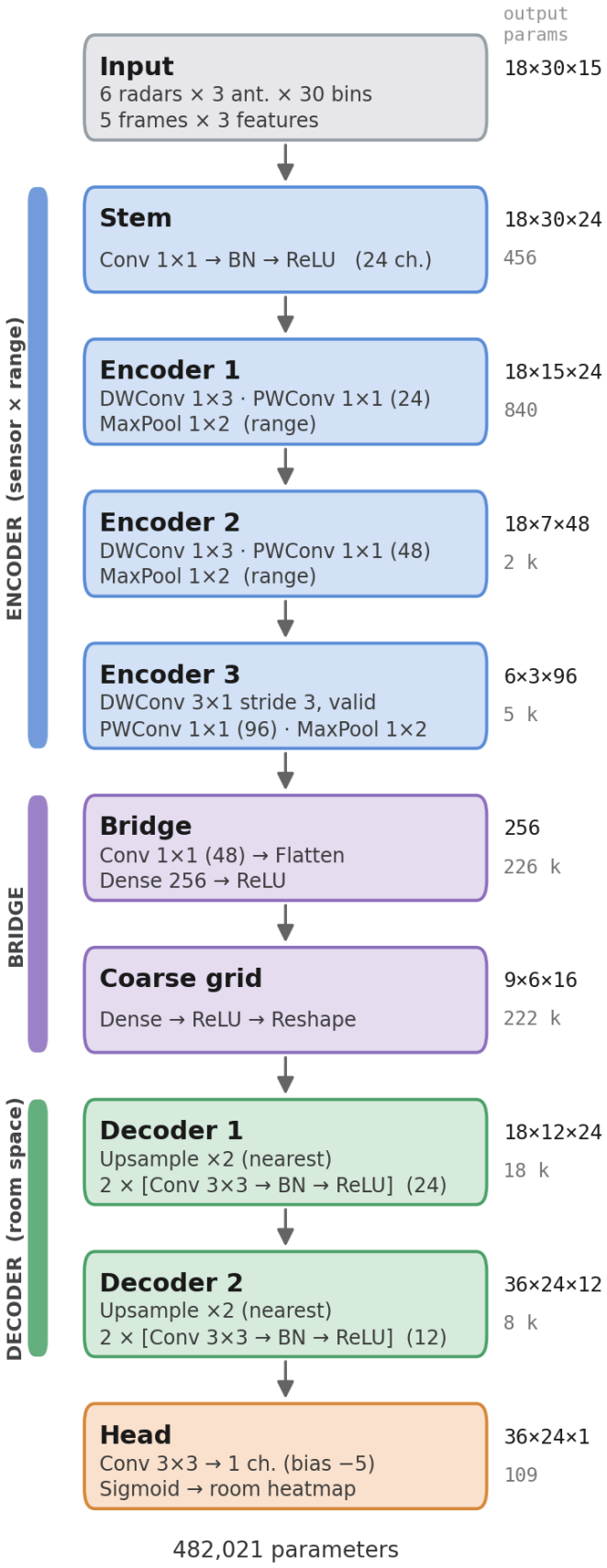


Fig. 4. Final TensorFlow Lite model, including the nominal parameter count.

the room, with rows as y coordinate and columns as x. Hence, the center of cell (r, c) represents the physical position:

$$(x, y) = ((c + 0.5) \cdot 0.2, \quad (r + 0.5) \cdot 0.2) \text{ m}. \tag{7}$$

Only one output channel is required because the model distinguishes person presence from background rather than among multiple object classes. The convolutional bias is initialized to -5, corresponding to a sigmoid probability of approximately 0.007. This prevents the untrained network from predicting foreground throughout the grid, where positive locations are extremely sparse.

To predict the target, we adopt the center-point representation introduced by CenterNet [2]. In the original method, each object is represented by the center of its bounding box and rendered onto a class-specific heatmap using an object-size-dependent Gaussian. We adapted this to the localization task by overlapping each valid position onto the room grid using a Gaussian with fixed standard deviation $\sigma = 1.5$ cells, corresponding to 0.3m. When multiple Gaussians overlap, their element-wise maximum is retained, as in CenterNet. Frames containing no people produce an all-zero target heatmap.

E. Training objective

Following Section 3 of [2], the heatmap is trained using the *penalty-reduced focal loss*:

$$\mathcal{L}_{\text{hm}} = -\frac{1}{N} \sum_{r,c} \mathcal{L}_{\text{r,c}}\left(Y_{\text{r,c}}, \hat{Y}_{\text{r,c}}\right)$$

$$\mathcal{L}_{\text{r,c}} = \begin{cases} \left(1 - \hat{Y}_{r,c}\right)^{\alpha} \log\left(\hat{Y}_{r,c}\right) & \text{if } Y_{r,c} = 1 \\ \left(1 - Y_{r,c}\right)^{\beta} \hat{Y}_{r,c}^{\alpha} \log\left(1 - \hat{Y}_{r,c}\right) & \text{else} \end{cases} \tag{8}$$

where $\alpha = 2$, $\beta = 4$, and N is the number of valid person centres, clamped to at least one for empty frames. The first term rewards correct centre predictions, while the second suppresses easy background cells. The $(1 - Y)^{\beta}$ factor reduces the negative penalty near a ground truth center, allowing smooth Gaussian responses (as in Section IV.D) rather than isolated valid cells. This differentiable objective learns center peaks converted into coordinates via Gaussian and evaluated through Hungarian matching. Our model *does not* directly optimize F1 at the 1-m radius.

F. Coasting and Quantization in Inference

At inference, candidate cells must be the maximum within their 3×3 neighbourhood and exceed the submitted confidence threshold of 0.30. Since the INT8 heatmap may contain plateaus where quantized values are identical where *quantized* values are identical, we discard all candidates within 0.30m of a stronger peak. The remaining peaks are *ranked by confidence* and *limited to the four people* supported by the task. A centroid weighted on confidence is applied over each peak's 3×3 neighbourhood, producing the final refined position.

To improve stationary target detections, i.e. decrease false negatives, we use previous predictions in inference. Specifically, two consecutive positive predictions confirm a "track". These confirmed detections may be repeated for *up to seven frames* (0.28 s) and associated within a 0.40 m gate. This *coasting* stage is causal; the submitted acquisition-mean background and centered temporal stack are not.

G. Edge Deployment Considerations

The final model uses full integer post-training quantization (PTQ), with INT8 input and output shapes of $1 \times 18 \times 30 \times 15$ and $1 \times 36 \times 24 \times 1$, respectively. The frozen binary occupies 521.52 KiB and its estimated TensorFlow Lite activation arena is 159.19 KiB, both below the 800 KiB and 300 KiB limits. The ESP32-S3 constraint checker reports no forbidden operations. Its static resource usage is summarized in Fig. 5.

The architecture has 480,245 trainable parameters (482,021 parameters in total) and 12.14 million MAC operations per frame. The two dense layers contain approximately 92% of the total parameters, whereas the convolutional decoder accounts for approximately 87% of the computation. The frozen model occupies 521.52 KiB and has an estimated TensorFlow Lite activation arena of 159.19 KiB, both within their respective static limits..

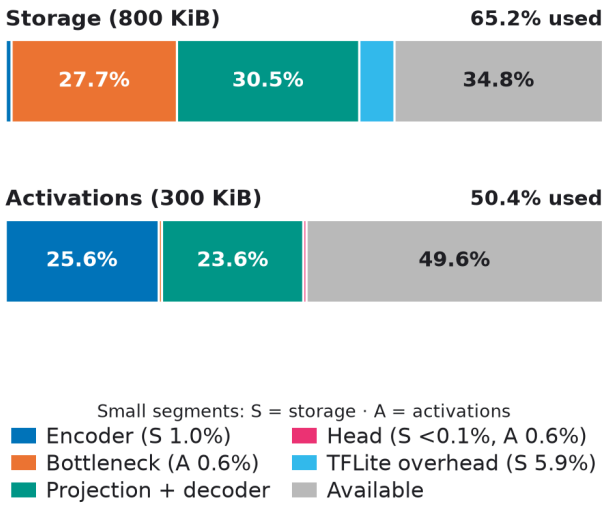


Fig. 5. Static model resource usage.

V. Experiments

We evaluate the two proposed models separately because they optimize different objectives. The submitted model is assessed primarily by localization quality under the official resource constraints. For the Causal EMA model, we also consider additional requirements imposed by online operation. Metrics use Hungarian matching with a 1-m acceptance radius. Comparisons are matched in split and configuration only where explicitly stated; separately trained, LOO, and Keras-screening results are reported as distinct protocols.

A. Accuracy-first model

The submitted full-INT8 pipeline was evaluated over all 15,000 frames (held-out windows `3` and `11`). It uses each acquisition's own mean, complete-acquisition percentile normalization, the centered five-frame stack, a 0.30 peak threshold, and the two-hit/seven-frame tracker described above.

TABLE I
Held-out accuracy of the submitted INT8 system.

Scope	Precision	Recall	F1 at 1-m
Pooled	0.9864	0.9641	0.9751
Window `3`	0.9839	0.9999	0.9918
Window `11`	0.9878	0.9462	0.9665

The pooled evaluation contains 43,384 true positives, 597 false positives, and 1,616 false negatives. Median, mean, 90th-percentile, and root-mean-square matched localization errors are 0.1391, 0.1718, 0.3334, and 0.2177 m, respectively. Tightening the matching radius gives F1 scores of 0.9388 at 0.5 m and 0.8075 at 0.25 m. Median error is 0.1674 m on window `3` and 0.1202 m on window `11`. The INT8 model matches the float32 reference in terms of performance, without loss, as shown in Fig. 6

a) *Different windows, different behaviour:*

Leave-one-window-out evaluation was performed over all 24 acquisition windows. In each fold, one window was used for testing, two for validation, and the remaining 21 for training. The pooled result is 0.9178 F1, with 0.9384 precision and 0.8981 recall. Windows `22` and `23` contain no people and are therefore degenerate for object-level F1; they should be interpreted as false-positive tests rather than ordinary localization folds.

Fig. 8 shows that performance is consistently high for most moving-person windows but deteriorates in recordings containing stationary or seated occupants. The per-person analysis in Fig. 9 is consistent with lower and less reliable recall at very small residual motion, although motion is not the only explanatory factor and window `15` is a notable counterexample. The sub-par performance on window `21` can be interpreted as evidence that almost completely stationary targets fail to be detected by the architecture, while already with 5 cm of residual motion the model is capable of detecting a person 50% of the times.

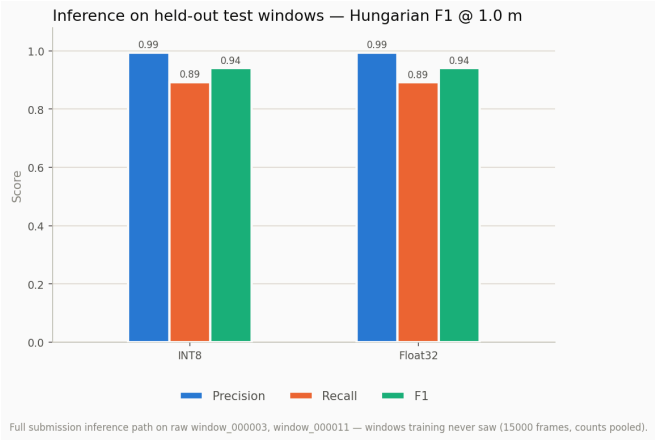


Fig. 6. Comparison between quantized and unquantized model.

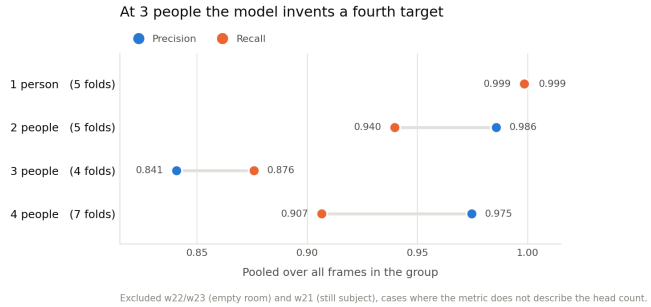


Fig. 7. F1 performance pooled by number of people.

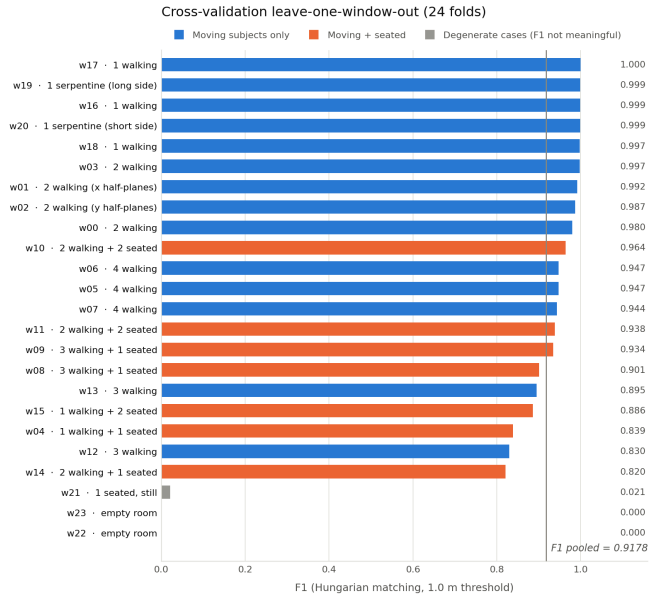


Fig. 8. F1 performance on cross validation, with description of content of each window.

All these results led to the decision of using coasting to improve recall on seated people, since they are detected correctly at least half the time (excluding window 21); this technique did not reduce performance on moving people in a meaningful way (refer to Table II).

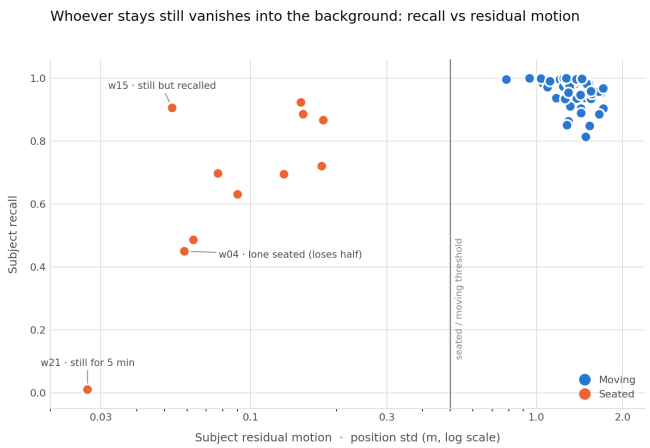


Fig. 9. Recall of seated people.

TABLE II
HELD-OUT COMPARISON OF THE RETAINED NON-COASTING BASELINE AND THE SUBMITTED COASTING CONFIGURATION ON WINDOWS 3 AND 11.

Configuration	Precision	Recall	F1	Median error
Without coasting	0.9946	0.8932	0.9412	0.1416 m
Submitted, with coasting	0.9864	0.9641	0.9751	0.1391 m

b) *Effect of coasting:*

Compared with the retained non-coasting baseline, the submitted configuration raises recall by 0.0709 and F1 by 0.0339, while precision decreases by only 0.0082. In terms of matched counts, it recovers 3,191 additional detections while introducing 378 additional false positives. Median localization error also decreases slightly, from 0.1416 to 0.1391 m. The gain therefore comes primarily from recovering intermittent missed detections rather than changing the localization accuracy of already detected targets.

This comparison demonstrates that the complete submitted configuration is stronger than the retained non-coasting baseline. Because the retained artifacts also differ in operating threshold or export history, it should be interpreted as a system-level comparison unless coasting is disabled and re-evaluated on the exact submitted binary with all other settings unchanged.

The paired INT8 and float32 inference comparison in Fig. 6 shows no material degradation after post-training quantization on the evaluated held-out run. Quantization is nevertheless justified primarily by the resource constraints: the submitted INT8 model occupies 521.52 KiB and has an estimated activation arena of 159.19 KiB.

B. Online variant

The submitted most-accurate preprocessing is not fully online: acquisition-mean subtraction and percentile normalization are performed on the complete recording, while the centered five-frame stack reads two future frames and introduces 80 ms of look-ahead. The Causal EMA background pre-processing removes these dependencies, by processing the stream from left to right using a stateful background estimate, a fixed training-derived normalization scale, and a past-only temporal buffer. Therefore, the prediction emitted at frame t depends only on observations with indices no greater than t.

Applying this streaming front end to the submitted weights obtains 0.9498 F1, with 0.9455 precision, 0.9542 recall, and 0.159 m median matched error on the same 15,000 held-out frames. Relative to the offline submitted system, this is a reduction of 0.0253 F1 and an increase of approximately 0.020 m in median error. This comparison quantifies the cost of removing complete-acquisition statistics and future-frame access; it does not isolate the effect of the EMA from the other front-end changes.

Host measurements require approximately 0.119 ms per frame for causal preprocessing and 0.111 ms for TFLite inference, well below the 40 ms frame period at 25 Hz. These measurements establish that the implementation can process a stream online on the evaluation host, but they are not measurements of ESP32-S3 latency.

a) *Scaling with temporal context:*

A separate single-seed Keras screening experiment evaluated causal temporal stacks of $K \in \{1, 5, 9\}$ frames. Every stack ends at the frame being predicted, so none uses future observations. The models share the same Causal EMA features, validation and test windows, seed, architecture settings, optimizer, and training budget.

TABLE III
SINGLE-SEED KERAS SCREENING OF CAUSAL TEMPORAL-CONTEXT LENGTH ON THE FIXED HELD-OUT PROTOCOL.

K	Past span	Validation F1	Test precision	Test recall	Test F1
1	0 ms	0.8141	0.9583	0.8380	0.8941
5	160 ms	0.8518	0.9825	0.8883	0.9330
9	320 ms	0.8599	0.9906	0.9318	0.9603

Increasing past context improves both validation and test performance. Relative to the single-frame model, the five-frame model raises test F1 by 0.0389, while the nine-frame model raises it by 0.0662. The improvement is driven mainly by recall, which increases from 0.8380 to 0.9318, while precision remains above 0.95 for every configuration.

At 25 Hz, $K = 5$ covers frames $t-4$ through t, or 160 ms of past context; $K = 9$ covers frames $t-8$ through t, or 320 ms. Missing startup history is zero-padded, allowing a prediction to be emitted for every incoming frame. The first fully populated stack becomes available after $K-1$ frames, which is a warm-up requirement rather than additional prediction look-ahead.

The $K = 9$ model was selected by validation F1, but it remains a Keras screening result. It has not replaced the submitted artifact and still requires INT8 export, constraint verification, and target-device timing.

C. Results

The submitted coasting configuration is the strongest retained held-out system, achieving 0.9751 F1 and 0.1391 m median matched error. Compared with the retained non-coasting baseline, it improves F1 by 3.39 percentage points, primarily through a 7.09-point increase in recall. The Causal

EMA front end provides a fully past-dependent alternative with 0.9498 F1, quantifying the accuracy cost of online operation. Finally, the causal temporal-context screening shows that longer past context improves recall and reaches 0.9603 Keras test F1 at $K = 9$, although this candidate is not yet an exported or target-validated replacement for the submitted model.

VI. Conclusion

The submitted accurate-variant INT8 model achieves 0.9751 F1 at the 1-m Hungarian-matching radius and 0.1391 m median matched error on held-out windows `3` and `11`; its 521.52 KiB model and 159.19 KiB estimated activation arena satisfy the stated static limits. Applying the causal EMA, fixed normalization, and past-only stack to the tested model removes the 80 ms look-ahead and obtains 0.9498 F1 on the same windows.

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